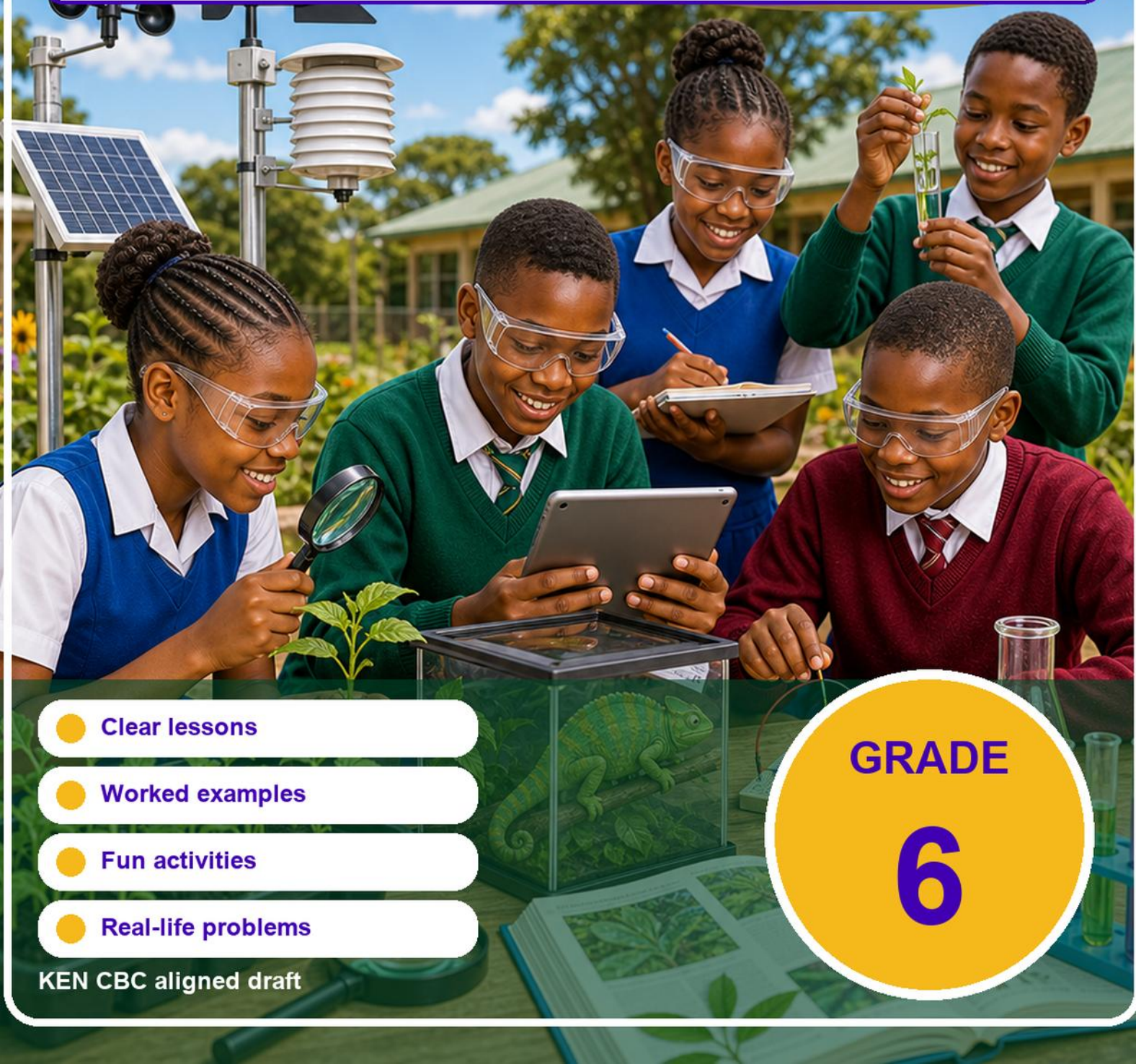


KITABU QUEST

Grade 6

SCIENCE AND TECHNOLOGY



- Clear lessons
- Worked examples
- Fun activities
- Real-life problems

KEN CBC aligned draft

GRADE

6

Title Page

KITABU QUEST Grade 6 Science and Technology

Kenya CBC content draft for review

Mascot: chameleon

This content-first draft was regenerated from Kitabu curriculum data using a topic-by-topic book plan. It is designed to help learners read, practise, discuss, create, and revise with confidence.

Cover artwork is intentionally deferred until all content has passed review and is marked ready-for-cover.

How To Use This Book

This book helps Grade 6 learners study Science and Technology through clear lessons, worked examples, activities, practice, and revision.

Use the inquiry questions to think first. Read the explanation. Try the guided example. Complete the activity. Answer the practice questions. Correct your work and ask for support where needed.

Observe, ask questions, test safely, record evidence, and explain findings.

Every unit has a local example, a misconception to avoid, success criteria, and a check for understanding.

Learning Skills In This Book

Each chapter builds knowledge, skills, values, creativity, communication, collaboration, critical thinking, digital literacy, and self-confidence.

Look for these page types: Learn, Example, Activity, Practice, Reflection, Home Link, and Review.

How to study well: read the goal first, underline new words, try the example before reading the answer, discuss with a partner, and correct your work after feedback. Do not skip the reflection questions. They help you notice what you understand and what needs more practice.

Table Of Contents

Use this table of contents to plan your study. Start with the first chapter, but return to earlier pages whenever you need revision.

1. 1.0 Living Things and their Environment
2. 2.0 Change of Matter State
3. 3.0 Force and Energy
4. 11.0 Living Things and their Environment

After each chapter, complete the chapter review before moving forward. At the end of the book, use the glossary, answer notes, and final project to revise the whole subject.

Chapter: Living Things and their Environment

In this chapter you will study Living Things and their Environment.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:

- 1.1 Fungi
- 1.3 Human circulatory system

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Fungi: Lesson Opener

Learning area: Fungi

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Fungi.

Curriculum link: This lesson follows the Grade 6 curriculum sequence and focuses on Fungi.

By the end of this lesson sequence, you should be able to:

- identify take a walk in the school compound and the common fungi adjacent environment to observe
- describe the observe and share with peers, importance of use print and non-print materials to search
- appreciate the discuss with peers

Inquiry questions:

- How is fungi important in nature?

Before reading, write one thing you already know and one question you want this lesson to answer.

Fungi: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Fungi.

Important idea: Fungi.

Environmental science connects observation with care for people and living things. Record place, date, conditions, and evidence. Then explain the pattern and propose a practical improvement that protects health, safety, food, water, or the environment.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 6, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Fungi |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows... |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Fungi

Explain the idea in your own words before attempting the activities.

Fungi: Worked Example

Investigation model for Fungi: Use field observation carefully.

Question: What evidence in the local environment shows the pattern or problem?

Example: A school garden with dry soil, wilting leaves, and cracks near the roots shows that plants need steady water and soil cover to reduce moisture loss.

Procedure:

1. Choose a safe observation point in the school compound, garden, home area, or community.
2. Record date, weather, place, and what you can see without disturbing living things.
3. Draw a labelled sketch or complete a three-column observation table.
4. Separate facts from explanations.
5. Suggest one practical improvement that protects health, safety, food, water, or the environment.

Conclusion frame: My observations show _____ because the evidence was _____.

Fungi: Activity

Activity for Fungi: Plan a safe observation or investigation.

Inquiry question: How is fungi important in nature?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Fungi: Activity And Practice

Practice for Fungi:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: identify take a walk in the school compound and the common fungi adjacent environment to observe. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: describe the observe and share with peers, importance of use print and non-print materials to search. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
3. Complete a science response for this outcome: appreciate the discuss with peers. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Fungi: Outcome Check

Outcome: identify take a walk in the school compound and the common fungi adjacent environment to observe

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Fungi, uses correct science vocabulary, and connects evidence to explanation.

Fungi: Outcome Check

Outcome: describe the observe and share with peers, importance of use print and non-print materials to search

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Fungi, uses correct science vocabulary, and connects evidence to explanation.

Fungi: Outcome Check

Outcome: appreciate the discuss with peers

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Fungi, uses correct science vocabulary, and connects evidence to explanation.

Human circulatory system: Lesson Opener

Learning area: Human circulatory system

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Human circulatory system.

Curriculum link: This lesson follows the Grade 6 curriculum sequence and focuses on Human circulatory system.

By the end of this lesson sequence, you should be able to:

- identify main parts of the human circulatory system
- describe functions of main parts of the human circulatory system
- outline the symptoms and prevention of common health conditions of the human circulatory

Inquiry questions:

- What is the human circulatory system made up of?
- What measures enhance a healthy human circulatory system?

Before reading, write one thing you already know and one question you want this lesson to answer.

Human circulatory system: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Human circulatory system.

Important idea: Human circulatory system.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 6, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Human circulatory system
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows...

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Human circulatory system
- Human
- circulatory
- system

Explain the idea in your own words before attempting the activities.

Human circulatory system: Worked Example

Investigation model for Human circulatory system: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:

1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
---	---	---
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Human circulatory system: Activity

Activity for Human circulatory system: Plan a safe observation or investigation.

Inquiry question: What is the human circulatory system made up of?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Human circulatory system: Activity And Practice

Practice for Human circulatory system:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: identify main parts of the human circulatory system. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: describe functions of main parts of the human circulatory system. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
3. Complete a science response for this outcome: outline the symptoms and prevention of common health conditions of the human circulatory. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Human circulatory system: Outcome Check

Outcome: identify main parts of the human circulatory system

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Human circulatory system, uses correct science vocabulary, and connects evidence to explanation.

Human circulatory system: Outcome Check

Outcome: describe functions of main parts of the human circulatory system

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Human circulatory system, uses correct science vocabulary, and connects evidence to explanation.

Human circulatory system: Outcome Check

Outcome: outline the symptoms and prevention of common health conditions of the human circulatory

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Human circulatory system, uses correct science vocabulary, and connects evidence to explanation.

Living Things and their Environment: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Chapter: Change of Matter State

In this chapter you will study Change of Matter State.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:

- 2.0 Change of Matter State

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Change of Matter State: Note: observe safety while heating substances to avoid: Lesson Opener

Learning area: Change of Matter State: Note: observe safety while heating substances to avoid

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Change of Matter State: Note: observe safety while heating substances to avoid.

Curriculum link: This lesson follows the Grade 6 curriculum sequence and focuses on Change of Matter State: Note: observe safety while heating substances to avoid.

By the end of this lesson sequence, you should be able to:

- Note: observe safety while heating substances to avoid fires and burns
- Project: Learners to make candles using waste candle wax or beeswax
- Learners to repair broken plastic containers.

Inquiry questions:

- How is change of state of matter important in day-to-day life?
- How does air pollution affect the environment?

Before reading, write one thing you already know and one question you want this lesson to answer.

Change of Matter State: Note: observe safety while heating substances to avoid: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Change of Matter State: Note: observe safety while heating substances to avoid.

Important idea: Change of Matter State: Note: observe safety while heating substances to avoid.

Materials can be described by observable properties such as state, texture, solubility, magnetism, colour, and reaction. Choose a separation or testing method because it matches a property, not because the method sounds familiar.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 6, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Change of Matter State: Note: observe safety while heating substances to avoid |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows... |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Change of Matter State: Note: observe safety while heating substances to avoid
- Change of Matter State
- Change
- Matter
- State
- Note
- observe
- safety

Explain the idea in your own words before attempting the activities.

Change of Matter State: Note: observe safety while heating substances to avoid: Worked Example

Investigation model for Change of Matter State: Note: observe safety while heating substances to avoid: Compare material properties.

Question: Which property helps us identify or separate the materials?

Example: A mixture of sand and salt can be separated because salt dissolves in water but sand does not. The sand can be filtered out, and salt can be recovered from the solution by evaporation.

Procedure:

1. List the materials and their visible properties.
2. Predict which property will be useful: size, solubility, magnetism, boiling point, texture, or colour change.
3. Choose a safe method that matches the property.
4. Record what happens before and after each step.
5. Explain why the method worked instead of only naming the method.

Conclusion frame: The useful property was _____, so the best method was _____.

Change of Matter State: Note: observe safety while heating substances to avoid: Activity

Activity for Change of Matter State: Note: observe safety while heating substances to avoid: Plan a safe observation or investigation.

Inquiry question: How is change of state of matter important in day-to-day life?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Change of Matter State: Note: observe safety while heating substances to avoid: Activity And Practice

Practice for Change of Matter State: Note: observe safety while heating substances to avoid:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: Note: observe safety while heating substances to avoid fires and burns. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: Project: Learners to make candles using waste candle wax or beeswax. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
3. Complete a science response for this outcome: Learners to repair broken plastic containers.. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Change of Matter State: Note: observe safety while heating substances to avoid: Outcome Check

Outcome: Note: observe safety while heating substances to avoid fires and burns

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Change of Matter State: Note: observe safety while heating substances to avoid, uses correct science vocabulary, and connects evidence to explanation.

Change of Matter State: Note: observe safety while heating substances to avoid: Outcome Check

Outcome: Project: Learners to make candles using waste candle wax or beeswax

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Change of Matter State: Note: observe safety while heating substances to avoid, uses correct science vocabulary, and connects evidence to explanation.

Change of Matter State: Note: observe safety while heating substances to avoid: Outcome Check

Outcome: Learners to repair broken plastic containers.

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Change of Matter State: Note: observe safety while heating substances to avoid, uses correct science vocabulary, and connects evidence to explanation.

Change of Matter State: components of air: Lesson Opener

Learning area: Change of Matter State: components of air

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Change of Matter State: components of air.

Curriculum link: This lesson follows the Grade 6 curriculum sequence and focuses on Change of Matter State: components of air.

By the end of this lesson sequence, you should be able to:

- identify the components of air
- outline uses of the different components of air
- explain the effects of air pollution in the environment

Inquiry questions:

- How does air pollution affect the environment?

Before reading, write one thing you already know and one question you want this lesson to answer.

Change of Matter State: components of air: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Change of Matter State: components of air.

Important idea: Change of Matter State: components of air.

Materials can be described by observable properties such as state, texture, solubility, magnetism, colour, and reaction. Choose a separation or testing method because it matches a property, not because the method sounds familiar.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 6, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Change of Matter State: components of air |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows... |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Change of Matter State: components of air
- Change of Matter State
- Change
- Matter
- State
- components

Explain the idea in your own words before attempting the activities.

Change of Matter State: components of air: Worked Example

Investigation model for Change of Matter State: components of air: Compare material properties.

Question: Which property helps us identify or separate the materials?

Example: A mixture of sand and salt can be separated because salt dissolves in water but sand does not. The sand can be filtered out, and salt can be recovered from the solution by evaporation.

Procedure:

1. List the materials and their visible properties.
2. Predict which property will be useful: size, solubility, magnetism, boiling point, texture, or colour change.
3. Choose a safe method that matches the property.
4. Record what happens before and after each step.
5. Explain why the method worked instead of only naming the method.

Conclusion frame: The useful property was _____, so the best method was _____.

Change of Matter State: components of air: Activity

Activity for Change of Matter State: components of air: Plan a safe observation or investigation.

Inquiry question: How does air pollution affect the environment?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Change of Matter State: components of air: Activity And Practice

Practice for Change of Matter State: components of air:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: identify the components of air. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: outline uses of the different components of air. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
3. Complete a science response for this outcome: explain the effects of air pollution in the environment. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Change of Matter State: components of air: Outcome Check

Outcome: identify the components of air

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Change of Matter State: components of air, uses correct science vocabulary, and connects evidence to explanation.

Change of Matter State: components of air: Outcome Check

Outcome: outline uses of the different components of air

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Change of Matter State: components of air, uses correct science vocabulary, and connects evidence to explanation.

Change of Matter State: components of air: Outcome Check

Outcome: explain the effects of air pollution in the environment

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Change of Matter State: components of air, uses correct science vocabulary, and connects evidence to explanation.

Change of Matter State: methods of reducing air pollution in the environment: Lesson Opener

Learning area: Change of Matter State: methods of reducing air pollution in the environment

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Change of Matter State: methods of reducing air pollution in the environment.

Curriculum link: This lesson follows the Grade 6 curriculum sequence and focuses on Change of Matter State: methods of reducing air pollution in the environment.

By the end of this lesson sequence, you should be able to:

- describe methods of reducing air pollution in the environment
- appreciate the need for clean air in day-to-day life.

Inquiry questions:

- How is change of state of matter important in day-to-day life?
- How does air pollution affect the environment?

Before reading, write one thing you already know and one question you want this lesson to answer.

Change of Matter State: methods of reducing air pollution in the environment: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Change of Matter State: methods of reducing air pollution in the environment.

Important idea: Change of Matter State: methods of reducing air pollution in the environment.

Materials can be described by observable properties such as state, texture, solubility, magnetism, colour, and reaction. Choose a separation or testing method because it matches a property, not because the method sounds familiar.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 6, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Change of Matter State: methods of reducing air pollution in the environment |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows... |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Change of Matter State: methods of reducing air pollution in the environment
- Change of Matter State
- Change
- Matter
- State
- methods
- reducing

Explain the idea in your own words before attempting the activities.

Change of Matter State: methods of reducing air pollution in the environment: Worked Example

Investigation model for Change of Matter State: methods of reducing air pollution in the environment: Compare material properties.

Question: Which property helps us identify or separate the materials?

Example: A mixture of sand and salt can be separated because salt dissolves in water but sand does not. The sand can be filtered out, and salt can be recovered from the solution by evaporation.

Procedure:

1. List the materials and their visible properties.
2. Predict which property will be useful: size, solubility, magnetism, boiling point, texture, or colour change.
3. Choose a safe method that matches the property.
4. Record what happens before and after each step.
5. Explain why the method worked instead of only naming the method.

Conclusion frame: The useful property was _____, so the best method was _____.

Change of Matter State: methods of reducing air pollution in the environment: Activity

Activity for Change of Matter State: methods of reducing air pollution in the environment: Plan a safe observation or investigation.

Inquiry question: How is change of state of matter important in day-to-day life?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Change of Matter State: methods of reducing air pollution in the environment: Activity And Practice

Practice for Change of Matter State: methods of reducing air pollution in the environment:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: describe methods of reducing air pollution in the environment. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: appreciate the need for clean air in day-to-day life.. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Change of Matter State: methods of reducing air pollution in the environment: Outcome Check

Outcome: describe methods of reducing air pollution in the environment

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Change of Matter State: methods of reducing air pollution in the environment, uses correct science vocabulary, and connects evidence to explanation.

Change of Matter State: methods of reducing air pollution in the environment: Outcome Check

Outcome: appreciate the need for clean air in day-to-day life.

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Change of Matter State: methods of reducing air pollution in the environment, uses correct science vocabulary, and connects evidence to explanation.

Change of Matter State: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Chapter: Force and Energy

In this chapter you will study Force and Energy.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:

- 3.1 Light
- 3.2 Levers as simple machines

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Light: movement of light through materials: Lesson Opener

Learning area: Light: movement of light through materials

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Light: movement of light through materials.

Curriculum link: This lesson follows the Grade 6 curriculum sequence and focuses on Light: movement of light through materials.

By the end of this lesson sequence, you should be able to:

- demonstrate the movement of light through materials
- draw ray diagrams of images formed in plane mirrors
- illustrate the formation of shadows and eclipses in nature

Inquiry questions:

- How does light travel?

Before reading, write one thing you already know and one question you want this lesson to answer.

Light: movement of light through materials: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Light: movement of light through materials.

Important idea: Light: movement of light through materials.

Physical science looks for patterns in energy, forces, and materials. Keep one condition changing at a time, observe the effect, and use measured evidence where possible. A correct explanation names the cause, the change observed, and the reason the change happened.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 6, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Light: movement of light through materials |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows... |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Light: movement of light through materials
- Light
- movement
- through
- materials

Explain the idea in your own words before attempting the activities.

Light: movement of light through materials: Worked Example

Investigation model for Light: movement of light through materials: Test one energy or circuit condition safely.

Question: What changes when one condition is changed?

Example: In a simple circuit, a bulb lights only when there is a complete path from one battery terminal, through the bulb, and back to the other terminal.

Procedure:

1. Name the variable you will change, such as number of cells, wire length, material, distance, or surface.
2. Name what you will observe or measure.
3. Keep the other conditions the same.
4. Record results in a table with at least three trials or comparisons.
5. Write a conclusion that mentions the pattern in the results.

Safety rule: Use only approved school materials and ask before connecting any electrical device.

Light: movement of light through materials: Activity

Activity for Light: movement of light through materials: Plan a safe observation or investigation.

Inquiry question: How does light travel?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Light: movement of light through materials: Activity And Practice

Practice for Light: movement of light through materials:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: demonstrate the movement of light through materials. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: draw ray diagrams of images formed in plane mirrors. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
3. Complete a science response for this outcome: illustrate the formation of shadows and eclipses in nature. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Light: movement of light through materials: Outcome Check

Outcome: demonstrate the movement of light through materials

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Light: movement of light through materials, uses correct science vocabulary, and connects evidence to explanation.

Light: movement of light through materials: Outcome Check

Outcome: draw ray diagrams of images formed in plane mirrors

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Light: movement of light through materials, uses correct science vocabulary, and connects evidence to explanation.

Light: movement of light through materials: Outcome Check

Outcome: illustrate the formation of shadows and eclipses in nature

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Light: movement of light through materials, uses correct science vocabulary, and connects evidence to explanation.

Light: formation of rainbow in nature: Lesson Opener

Learning area: Light: formation of rainbow in nature

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Light: formation of rainbow in nature.

Curriculum link: This lesson follows the Grade 6 curriculum sequence and focuses on Light: formation of rainbow in nature.

By the end of this lesson sequence, you should be able to:

- describe the formation of rainbow in nature
- appreciate the importance of movement light in everyday life.

Inquiry questions:

- How does light travel?

Before reading, write one thing you already know and one question you want this lesson to answer.

Light: formation of rainbow in nature: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Light: formation of rainbow in nature.

Important idea: Light: formation of rainbow in nature.

Physical science looks for patterns in energy, forces, and materials. Keep one condition changing at a time, observe the effect, and use measured evidence where possible. A correct explanation names the cause, the change observed, and the reason the change happened.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 6, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Light: formation of rainbow in nature |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows... |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Light: formation of rainbow in nature
- Light
- formation
- rainbow
- nature

Explain the idea in your own words before attempting the activities.

Light: formation of rainbow in nature: Worked Example

Investigation model for Light: formation of rainbow in nature: Test one energy or circuit condition safely.

Question: What changes when one condition is changed?

Example: In a simple circuit, a bulb lights only when there is a complete path from one battery terminal, through the bulb, and back to the other terminal.

Procedure:

1. Name the variable you will change, such as number of cells, wire length, material, distance, or surface.
2. Name what you will observe or measure.
3. Keep the other conditions the same.
4. Record results in a table with at least three trials or comparisons.
5. Write a conclusion that mentions the pattern in the results.

Safety rule: Use only approved school materials and ask before connecting any electrical device.

Light: formation of rainbow in nature: Activity

Activity for Light: formation of rainbow in nature: Plan a safe observation or investigation.

Inquiry question: How does light travel?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Light: formation of rainbow in nature: Activity And Practice

Practice for Light: formation of rainbow in nature:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: describe the formation of rainbow in nature. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: appreciate the importance of movement light in everyday life.. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Light: formation of rainbow in nature: Outcome Check

Outcome: describe the formation of rainbow in nature

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Light: formation of rainbow in nature, uses correct science vocabulary, and connects evidence to explanation.

Light: formation of rainbow in nature: Outcome Check

Outcome: appreciate the importance of movement light in everyday life.

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Light: formation of rainbow in nature, uses correct science vocabulary, and connects evidence to explanation.

Levers as simple machines: common levers used in day- to-day life: Lesson Opener

Learning area: Levers as simple machines: common levers used in day- to-day life

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Levers as simple machines: common levers used in day- to-day life.

Curriculum link: This lesson follows the Grade 6 curriculum sequence and focuses on Levers as simple machines: common levers used in day- to-day life.

By the end of this lesson sequence, you should be able to:

- identify common levers used in day- to-day life
- describe parts of a lever as used in making work easier
- classify levers into the three classes

Inquiry questions:

- How are levers used in day to day life?

Before reading, write one thing you already know and one question you want this lesson to answer.

Levers as simple machines: common levers used in day- to-day life: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Levers as simple machines: common levers used in day- to-day life.

Important idea: Levers as simple machines: common levers used in day- to-day life.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 6, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Levers as simple machines: common levers used in day- to-day life
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows...

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Levers as simple machines: common levers used in day- to-day life
- Levers as simple machines
- Levers
- simple
- machines
- common

Explain the idea in your own words before attempting the activities.

Levers as simple machines: common levers used in day- to-day life: Worked Example

Investigation model for Levers as simple machines: common levers used in day- to-day life: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:

1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
---	---	---
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Levers as simple machines: common levers used in day- to-day life: Activity

Activity for Levers as simple machines: common levers used in day- to-day life: Plan a safe observation or investigation.

Inquiry question: How are levers used in day to day life?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Levers as simple machines: common levers used in day- to-day life: Activity And Practice

Practice for Levers as simple machines: common levers used in day- to-day life:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: identify common levers used in day- to-day life. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: describe parts of a lever as used in making work easier. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
3. Complete a science response for this outcome: classify levers into the three classes. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Levers as simple machines: common levers used in day- to-day life: Outcome Check

Outcome: identify common levers used in day- to-day life

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Levers as simple machines: common levers used in day- to-day life, uses correct science vocabulary, and connects evidence to explanation.

Levers as simple machines: common levers used in day- to-day life: Outcome Check

Outcome: describe parts of a lever as used in making work easier

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Levers as simple machines: common levers used in day- to-day life, uses correct science vocabulary, and connects evidence to explanation.

Levers as simple machines: common levers used in day- to-day life: Outcome Check

Outcome: classify levers into the three classes

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Levers as simple machines: common levers used in day- to-day life, uses correct science vocabulary, and connects evidence to explanation.

Levers as simple machines: use of levers in making work easier: Lesson Opener

Learning area: Levers as simple machines: use of levers in making work easier

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Levers as simple machines: use of levers in making work easier.

Curriculum link: This lesson follows the Grade 6 curriculum sequence and focuses on Levers as simple machines: use of levers in making work easier.

By the end of this lesson sequence, you should be able to:

- demonstrate the use of levers in making work easier
- appreciate the use of levers in making work easier.

Inquiry questions:

- How are levers used in day to day life?

Before reading, write one thing you already know and one question you want this lesson to answer.

Levers as simple machines: use of levers in making work easier: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Levers as simple machines: use of levers in making work easier.

Important idea: Levers as simple machines: use of levers in making work easier.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 6, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Levers as simple machines: use of levers in making work easier
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows...

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Levers as simple machines: use of levers in making work easier
- Levers as simple machines
- Levers
- simple
- machines
- making
- easier

Explain the idea in your own words before attempting the activities.

Levers as simple machines: use of levers in making work easier: Worked Example

Investigation model for Levers as simple machines: use of levers in making work easier: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:

1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Levers as simple machines: use of levers in making work easier: Activity

Activity for Levers as simple machines: use of levers in making work easier: Plan a safe observation or investigation.

Inquiry question: How are levers used in day to day life?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Levers as simple machines: use of levers in making work easier: Activity And Practice

Practice for Levers as simple machines: use of levers in making work easier:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: demonstrate the use of levers in making work easier. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: appreciate the use of levers in making work easier.. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Levers as simple machines: use of levers in making work easier: Outcome Check

Outcome: demonstrate the use of levers in making work easier

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Levers as simple machines: use of levers in making work easier, uses correct science vocabulary, and connects evidence to explanation.

Levers as simple machines: use of levers in making work easier: Outcome Check

Outcome: appreciate the use of levers in making work easier.

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Levers as simple machines: use of levers in making work easier, uses correct science vocabulary, and connects evidence to explanation.

Force and Energy: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Chapter: Living Things and their Environment

In this chapter you will study Living Things and their Environment.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:

- 1.2 Invertebrates

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Invertebrates: common invertebrates in the environment: Lesson Opener

Learning area: Invertebrates: common invertebrates in the environment

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Invertebrates: common invertebrates in the environment.

Curriculum link: This lesson follows the Grade 6 curriculum sequence and focuses on Invertebrates: common invertebrates in the environment.

By the end of this lesson sequence, you should be able to:

- identify common invertebrates in the environment
- practise precautions in handling invertebrates
- describe the general characteristics of invertebrates

Inquiry questions:

- What are the common features of invertebrates?

Before reading, write one thing you already know and one question you want this lesson to answer.

Invertebrates: common invertebrates in the environment: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Invertebrates: common invertebrates in the environment.

Important idea: Invertebrates: common invertebrates in the environment.

Environmental science connects observation with care for people and living things. Record place, date, conditions, and evidence. Then explain the pattern and propose a practical improvement that protects health, safety, food, water, or the environment.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 6, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Invertebrates: common invertebrates in the environment
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows...

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Invertebrates: common invertebrates in the environment
- Invertebrates
- common
- environment

Explain the idea in your own words before attempting the activities.

Invertebrates: common invertebrates in the environment: Worked Example

Investigation model for Invertebrates: common invertebrates in the environment: Use field observation carefully.

Question: What evidence in the local environment shows the pattern or problem?

Example: A school garden with dry soil, wilting leaves, and cracks near the roots shows that plants need steady water and soil cover to reduce moisture loss.

Procedure:

1. Choose a safe observation point in the school compound, garden, home area, or community.
2. Record date, weather, place, and what you can see without disturbing living things.
3. Draw a labelled sketch or complete a three-column observation table.
4. Separate facts from explanations.
5. Suggest one practical improvement that protects health, safety, food, water, or the environment.

Conclusion frame: My observations show _____ because the evidence was _____.

Invertebrates: common invertebrates in the environment: Activity

Activity for Invertebrates: common invertebrates in the environment: Plan a safe observation or investigation.

Inquiry question: What are the common features of invertebrates?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Invertebrates: common invertebrates in the environment: Activity And Practice

Practice for Invertebrates: common invertebrates in the environment:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: identify common invertebrates in the environment. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: practise precautions in handling invertebrates. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
3. Complete a science response for this outcome: describe the general characteristics of invertebrates. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Invertebrates: common invertebrates in the environment: Outcome Check

Outcome: identify common invertebrates in the environment

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Invertebrates: common invertebrates in the environment, uses correct science vocabulary, and connects evidence to explanation.

Invertebrates: common invertebrates in the environment: Outcome Check

Outcome: practise precautions in handling invertebrates

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Invertebrates: common invertebrates in the environment, uses correct science vocabulary, and connects evidence to explanation.

Invertebrates: common invertebrates in the environment: Outcome Check

Outcome: describe the general characteristics of invertebrates

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Invertebrates: common invertebrates in the environment, uses correct science vocabulary, and connects evidence to explanation.

Invertebrates: outline the economic importance of invertebrates: Lesson Opener

Learning area: Invertebrates: outline the economic importance of invertebrates

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Invertebrates: outline the economic importance of invertebrates.

Curriculum link: This lesson follows the Grade 6 curriculum sequence and focuses on Invertebrates: outline the economic importance of invertebrates.

By the end of this lesson sequence, you should be able to:

- outline the economic importance of invertebrates
- appreciate the importance of invertebrates in nature.

Inquiry questions:

- What are the common features of invertebrates?

Before reading, write one thing you already know and one question you want this lesson to answer.

Invertebrates: outline the economic importance of invertebrates: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Invertebrates: outline the economic importance of invertebrates.

Important idea: Invertebrates: outline the economic importance of invertebrates.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 6, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Invertebrates: outline the economic importance of invertebrates
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows...

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Invertebrates: outline the economic importance of invertebrates
- Invertebrates
- outline
- economic

Explain the idea in your own words before attempting the activities.

Invertebrates: outline the economic importance of invertebrates: Worked Example

Investigation model for Invertebrates: outline the economic importance of invertebrates: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:

1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
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Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Invertebrates: outline the economic importance of invertebrates: Activity

Activity for Invertebrates: outline the economic importance of invertebrates: Plan a safe observation or investigation.

Inquiry question: What are the common features of invertebrates?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Invertebrates: outline the economic importance of invertebrates: Activity And Practice

Practice for Invertebrates: outline the economic importance of invertebrates:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: outline the economic importance of invertebrates. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: appreciate the importance of invertebrates in nature.. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Invertebrates: outline the economic importance of invertebrates: Outcome Check

Outcome: outline the economic importance of invertebrates

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Invertebrates: outline the economic importance of invertebrates, uses correct science vocabulary, and connects evidence to explanation.

Invertebrates: outline the economic importance of invertebrates: Outcome Check

Outcome: appreciate the importance of invertebrates in nature.

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Invertebrates: outline the economic importance of invertebrates, uses correct science vocabulary, and connects evidence to explanation.

Living Things and their Environment: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Fungi: Review Clinic 1

Use this review clinic to strengthen a real unit from the book.

Practice context: home or school example connected to Fungi.

1. Re-read the lesson on Fungi and underline the main idea.
2. Write a five-sentence or five-step summary.
3. Make one question that tests knowledge.
4. Make one question that tests skill, reasoning, or application.
5. Solve or answer both questions.
6. Check your work against this criterion: observation is specific.
7. Exchange your questions with a partner and discuss corrections.

Challenge: Describe a real-life task from your home, school, farm, market, field, library, or community that uses this idea.

Human circulatory system: Review Clinic 2

Use this review clinic to strengthen a real unit from the book.

Practice context: county or community example connected to Human circulatory system.

1. Re-read the lesson on Human circulatory system and underline the main idea.
2. Write a five-sentence or five-step summary.
3. Make one question that tests knowledge.
4. Make one question that tests skill, reasoning, or application.
5. Solve or answer both questions.
6. Check your work against this criterion: safety is considered.
7. Exchange your questions with a partner and discuss corrections.

Challenge: Describe a real-life task from your home, school, farm, market, field, library, or community that uses this idea.

Change of Matter State: methods of reducing air pollution in the environment: Review Clinic 3

Use this review clinic to strengthen a real unit from the book.

Practice context: partner discussion about Change of Matter State: methods of reducing air pollution in the environment.

1. Re-read the lesson on Change of Matter State: methods of reducing air pollution in the environment and underline the main idea.
2. Write a five-sentence or five-step summary.
3. Make one question that tests knowledge.
4. Make one question that tests skill, reasoning, or application.
5. Solve or answer both questions.
6. Check your work against this criterion: conclusion uses evidence.
7. Exchange your questions with a partner and discuss corrections.

Challenge: Describe a real-life task from your home, school, farm, market, field, library, or community that uses this idea.

Light: formation of rainbow in nature: Review Clinic 4

Use this review clinic to strengthen a real unit from the book.

Practice context: small project or observation linked to Light: formation of rainbow in nature.

1. Re-read the lesson on Light: formation of rainbow in nature and underline the main idea.
2. Write a five-sentence or five-step summary.
3. Make one question that tests knowledge.
4. Make one question that tests skill, reasoning, or application.
5. Solve or answer both questions.
6. Check your work against this criterion: observation is specific.
7. Exchange your questions with a partner and discuss corrections.

Challenge: Describe a real-life task from your home, school, farm, market, field, library, or community that uses this idea.

Glossary

Use this glossary to revise important words in Science and Technology.

- Fungi: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Human circulatory system: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Change of Matter State: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Light: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Levers as simple machines: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Invertebrates: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.

Add five more words from your lessons, then write a meaning and one correct example sentence or worked example for each word.

Answer And Teacher Notes

A strong Science and Technology answer should:

- answer the exact question asked;
- separate observation from explanation;
- name materials, variables, safety checks, or data sources where needed;
- use labelled diagrams, tables, or step-by-step procedures when useful;
- draw a conclusion from evidence, not from guessing.

For investigations, good answers include a question, method, results, conclusion, and one limit or improvement. For technology tasks, good answers explain the design choice and how it solves the problem safely.

Answer-key guide: a complete science answer names the concept, shows evidence in a labelled sketch/table/model, explains the pattern, includes safety where relevant, and ends with a conclusion beginning from the evidence. Wrong answers often mix observation with opinion or leave out variables.

Rubric for self-checking:

- 4: The answer addresses the exact question, uses subject vocabulary correctly, gives evidence or working, and includes a useful check or correction.
- 3: The main answer is correct but one part needs stronger evidence, clearer working, better labels, or a fuller explanation.
- 2: Some understanding is shown, but the answer is incomplete, too general, or hard to follow.
- 1: The answer is guessed, off topic, or missing the evidence needed for the task.

Correction routine: reread the command word, underline the key data or evidence, improve one weak step, and write one sentence explaining why the final answer is reasonable.

Final Project

Create a final Science and Technology project.

Your project must include:

1. A clear question, problem, or design need.
2. Evidence from at least three lessons.
3. A safe method, labelled diagram, data table, flow chart, model, or prototype plan.
4. A conclusion that explains what the evidence shows.
5. One safety note, one limitation, and one improvement.

Present your project to a classmate, teacher, parent, or study group.